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Introduction to Oracle Objects

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Object Type

Object Type *person_typ*

Attributes	Methods
idno	get_idno
first_name	display_details
last_name	
email	
phone	

Object

idno:	65
first_name:	Verna
last_name:	Mills
email:	vmills@oracle.com
phone:	1-800-555-4412

Object

idno:	101
first_name:	John
last_name:	Smith
email:	jsmith@oracle.com
phone:	1-800-555-1212

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Creating the Type

```
CREATE TYPE person_typ AS OBJECT (
  idno           NUMBER,
  first_name     VARCHAR2(20),
  last_name      VARCHAR2(25),
  email          VARCHAR2(25),
  phone          VARCHAR2(20),
  MAP MEMBER FUNCTION get_idno RETURN NUMBER,
  MEMBER PROCEDURE display_details
( SELF IN OUT NOCOPY person_typ ));
```

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Creating the Type

```
CREATE TYPE BODY person_typ AS
  MAP MEMBER FUNCTION get_idno RETURN NUMBER IS
  BEGIN
    RETURN idno;
  END;
  MEMBER PROCEDURE display_details
   ( SELF IN OUT NOCOPY person_typ ) IS
  BEGIN
    DBMS_OUTPUT.PUT_LINE(TO_CHAR(idno) || ' '
      || first_name || ' ' || last_name);
    DBMS_OUTPUT.PUT_LINE(email || ' '  || phone);
  END;
END;
```

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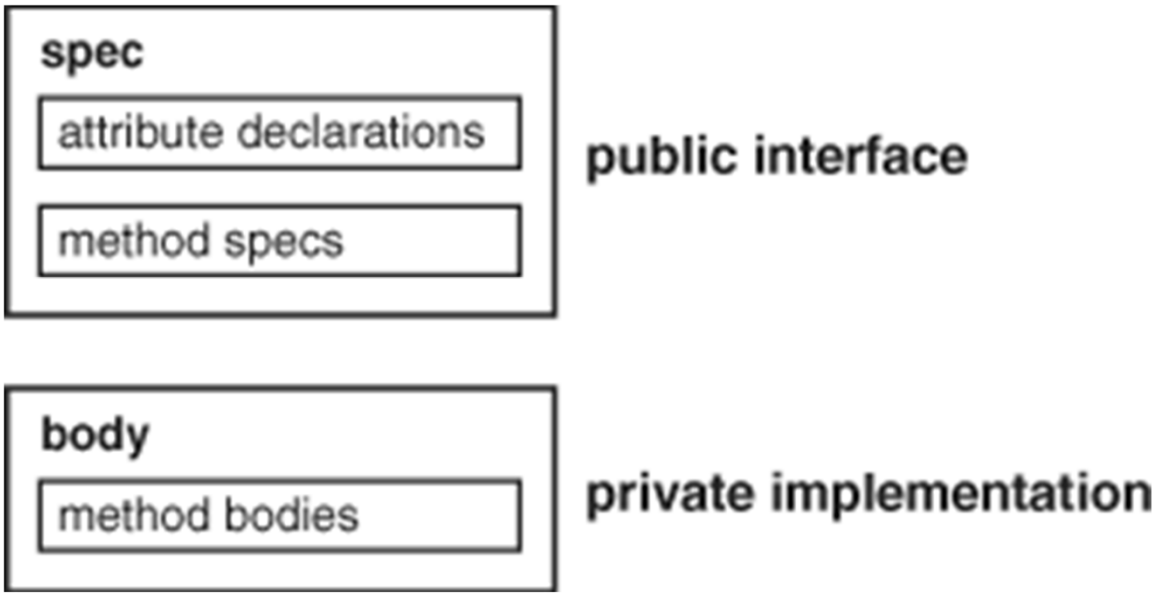
Object Attributes and Methods



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Creating the Object Table

```
CREATE TABLE person_obj_table OF person_typ;

 INSERT INTO person_obj_table
  VALUES (person_typ(101, 'John', 'Smith'
    , 'jsmith@oracle.com', '1-800-555-1212') );

SELECT VALUE(p) FROM person_obj_table p
       WHERE p.last_name = 'Smith';
```

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Using SELECT INTO

```
DECLARE
  person person_typ;
BEGIN
  SELECT VALUE(p)
     INTO person
     FROM person_obj_table p
     WHERE p.idno = 101;
  person.display_details();
END;
```

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Using REF to the emp_person_typ Object

```
CREATE TYPE emp_person_typ AS OBJECT (
  name     VARCHAR2(30),
  manager  REF emp_person_typ );

CREATE TABLE emp_person_obj_table OF emp_person_typ;

CREATE TABLE contacts_ref (
  contact_ref   REF person_typ SCOPE IS person_obj_table,
  contact_date  DATE );
```

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Subtype and Hierarchy

```
CREATE TYPE person_t AS OBJECT
  (name VARCHAR2(100) , ssn NUMBER)
   NOT FINAL;


CREATE TYPE employee_t UNDER person_t
   (department_id NUMBER, salary NUMBER) NOT FINAL;


CREATE TYPE part_time_emp_t UNDER employee_t
   (num_hrs NUMBER);
```

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